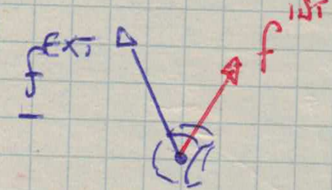
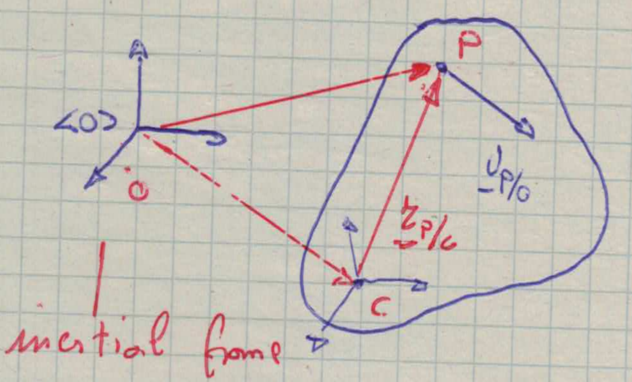
I CARDINAL EQUATIONS (NEWTON EQUATION)

Consider a body $\mathcal{B}$ (*)



functions of P



$$\frac{d_O}{dt}\underbrace{\underline{v}_{P/O}}_{\underline{v}_{C/O}+\underline{\omega}_{b/O}\times\underline{r}_{P/C}}\,dm = \underline{f}^{INT} + \underline{f}^{EXT}$$

$$\int_V \frac{d_O}{dt}\underline{v}_{P/O}\,\underbrace{\rho\, dV}_{dm} = \int_V \left(\underline{f}^{EXT}+\underline{f}^{INT}\right) dV = \underbrace{\int_V \underline{f}^{EXT}\, dV}_{\underline{F}^{EXT}} + \underbrace{\int_V \underline{f}^{INT}\, dV}_{\emptyset}$$

Then

$$\int_V \underbrace{\frac{d_O}{dt}\left[\underline{v}_{C/O} + \underline{\omega}_{b/O}\times\underline{r}_{P/C}\right]}_{\underline{a}_{C/O} + \dot{\underline{\omega}}_{b/O}\times\underline{r}_{P/C} + \underline{\omega}_{b/O}\times\frac{d_O}{dt}\underline{r}_{P/C}}\rho\, dV = \underline{F}^{EXT}$$

$$\underbrace{\frac{d_O}{dt}\underline{r}_{P/C}}_{\cancel{\frac{d_b}{dt}\underline{r}_{P/C}}_{\emptyset} + \underline{\omega}_{b/O}\times\underline{r}_{P/C}}$$

Then

$$\underline{a}_{C/O}\underbrace{\int_V \rho\, dV}_{m} + \dot{\underline{\omega}}_{b/O}\times\underbrace{\cancel{\int_V \underline{r}_{P/C}\,\rho\, dV}}_{m\underline{r}_{C/C}=0} + \underline{\omega}_{b/O}\times\left[\underline{\omega}_{b/O}\underbrace{\cancel{\int_V \underline{r}_{P/C}\,\rho\, dV}}_{m\underline{r}_{C/C}=0}\right] = \underline{F}^{EXT}$$

$$m\,\underline{a}_{C/O} = \underline{F}^{EXT} \longleftarrow \text{I dynamic cardinal eq.}$$

$$\underline{0} = \underline{F}^{EXT} \longleftarrow \text{I static cardinal eq.}$$

(*) We assume m to be constant.

NOTE $\underline{F}^{EXT}$ is the total contribution of the (external) forces acting on $\mathcal{B}$

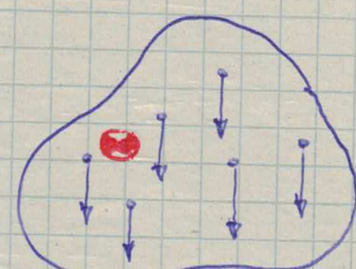
NOTE The static I cardinal eq. refers to the $\mathcal{B}$ equilibrium state corresponding to $\underline{v}_{c/o} \equiv \underline{\omega}_{b/o} \equiv \underline{0}$

For the dynamic case the I cardinal eq can be projected onto frame <0> leading to the following equations

$$m\,{}^{0}\underline{a}_{c/o} = {}^{0}\underline{F}^{EXT} \iff \begin{cases} m \dfrac{d}{dt}\,{}^{0}\underline{v}_{c/o} = {}^{0}\underline{F}^{EXT} \\ \dfrac{d}{dt}\,{}^{0}\underline{r}_{c/o} = {}^{0}\underline{v}_{c/o} \end{cases} \quad (*)$$

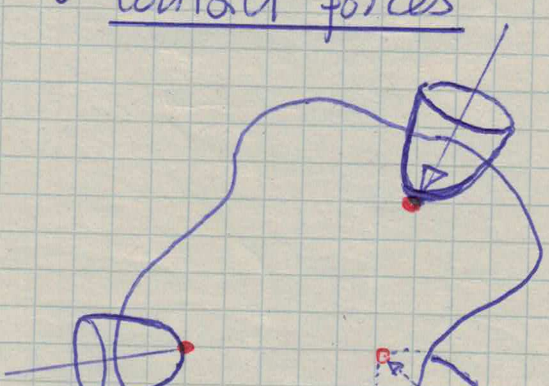
Examples of external forces

- Gravity



$$\underline{F}_g^{EXT} = \int_V \underline{g}\,dm = m\underline{g}$$

- Contact forces



Contact models:

point contacts (w/ or w/o friction)

$$\underline{F}_c^{EXT} = \Sigma_i\, \underline{f}_{c_i}$$

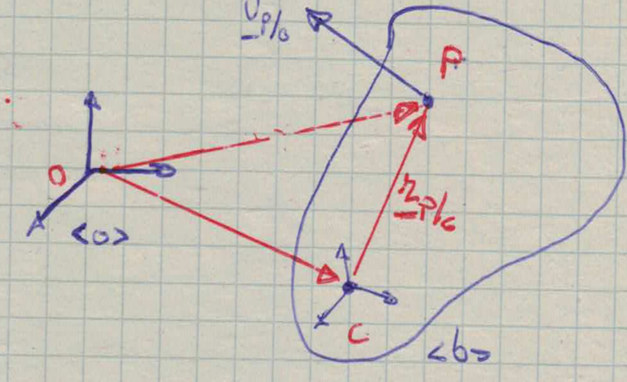
distributed contacts

$$\underline{F}_c^{EXT} = \int_S \underline{f}_{c_i}\, ds$$

Tactile sensors required (often)

(*) 6 first order diff. eq.s. (state form)

II CARDINAL EQUATIONS (EULER EQUATION)



Consider

$$\frac{d_o}{dt}\underline{v}_{P/O}\,dm = \underline{f}^{INT} + \underline{f}^{EXT}$$

then compute the momentum about point C

$$\underbrace{\underline{r}_{P/C} \times \left(\frac{d_o}{dt}\underline{v}_{P/O}\right) dm}_{\circledast} = \underline{r}_{P/C} \times \left[\underline{f}^{INT} + \underline{f}^{EXT}\right] \Rightarrow$$

$$= \underbrace{\underline{r}_{P/C} \times \underline{f}^{INT}}_{\underline{m}^{INT}_{P/C}} + \underbrace{\underline{r}_{P/C} \times \underline{f}^{EXT}}_{\underline{m}^{EXT}_{P/C}}$$

Consider the quantity:

$$\frac{d_o}{dt}\left(\underline{r}_{P/C} \times \underline{v}_{P/O}\right) = \left(\frac{d_o}{dt}\underline{r}_{P/C}\right) \times \underline{v}_{P/O} + \underbrace{\underline{r}_{P/C} \times \frac{d_o}{dt}\underline{v}_{P/O}}_{\circledast}$$

$$\frac{d_o}{dt}\left(\underline{r}_{P/O} - \underline{r}_{C/O}\right) = \underline{v}_{P/O} - \underline{v}_{C/O}$$

Then

$$\circledast = \underline{r}_{P/C} \times \frac{d_o}{dt}\underline{v}_{P/O} = \frac{d_o}{dt}\left(\underline{r}_{P/C} \times \underline{v}_{P/O}\right) + \underline{v}_{C/O} \times \underline{v}_{P/O}$$

<u>NOTE</u> These steps are useful since $\circledast$ must be integrated and time derivative inside the integral is annoying.

Then

$$\int_V \left[\underline{r}_{P/c} \times \frac{d_o}{dt} \underline{v}_{P/o} \right] dm = \underbrace{\int_V \left[\underline{r}_{P/c} \times \underline{f}^{INT} \right] dV}_{\underline{M}^{INT}} + \underbrace{\int_V \left[\underline{r}_{P/c} \times \underline{f}^{EXT} \right] dV}_{\underline{M}^{EXT}}$$

can be also re-written as

$$\int_V \frac{d_o}{dt} \left(\underline{r}_{P/c} \times \underline{v}_{P/o} \right) \rho\, dV + \int_V \left(\underline{v}_{c/o} \times \underline{v}_{P/o} \right) \rho\, dV = \underline{M}^{EXT} + \underline{M}^{INT}$$

then

$$\frac{d_o}{dt} \int_V \underline{r}_{P/c} \times \left[\underline{v}_{c/o} + \underline{\omega}_{b/o} \times \underline{r}_{P/c} \right] \rho\, dV + \underline{v}_{c/o} \times \int_V \underline{v}_{P/o} \rho\, dV =$$

$$\underline{v}_{c/o} + \underline{\omega}_{b/o} \times \underline{r}_{P/c}$$

$$= \frac{d_o}{dt} \left[\Big(\underbrace{\int_V \underline{r}_{P/c} \rho\, dV}_{m \underline{r}_{c/c} = \underline{0}} \Big) \times \underline{v}_{c/o} + \underbrace{\int_V \underline{r}_{P/c} \times \left[\underline{\omega}_{b/o} \times \underline{r}_{P/c} \right] \rho\, dV}_{I_c(\underline{\omega}_{b/o})} \right] +$$

$$+ \underline{v}_{c/o} \times \left[m \underline{v}_{c/o} + \underline{\omega}_{b/o} \times \underbrace{\int_V \underline{r}_{P/c} \rho\, dV}_{m \underline{r}_{c/c} = \underline{0}} \right]$$

$$\underline{v}_{c/o} \times m \underline{v}_{c/o} = \underline{0}$$

Finally we obtain the equality

$$\frac{d_o}{dt} I_c(\underline{\omega}_{b/o}) = \underline{M}^{EXT} + \underline{M}^{INT} \qquad (*)$$

NOTE Internal forces are "balanced" i.e.

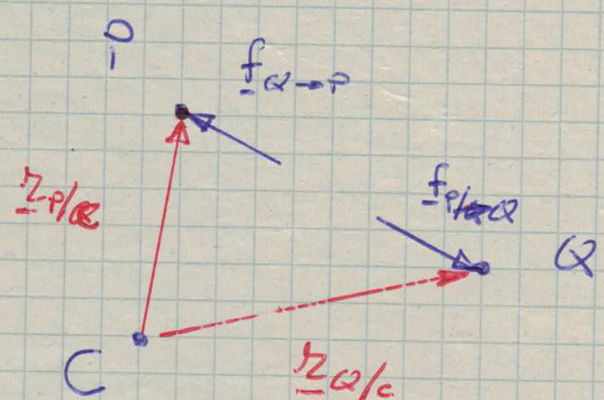
$$\underline{f}_P^{INT} = \int_V \underline{f}_{Q\to P} \, dV$$

internal force acting on P

$$\forall \, \underline{f}_{Q\to P}, \; \exists \; \underline{f}_{P\to Q} = -\underline{f}_{Q\to P} \quad \text{s.t.}$$

$$(P-Q) \times \underline{f}_{P\to Q} = 0$$

For each $P, Q \in \mathcal{B}$ we have



$$\underline{r}_{Q/C} \times \underline{f}_{P\to Q} + \underline{r}_{P/C} \times \underline{f}_{Q\to P} =$$

$$= \underline{r}_{Q/C} \times \underline{f}_{P\to Q} - \underline{r}_{P/C} \times \underline{f}_{P\to Q} =$$

$$= (\underline{r}_{Q/C} - \underline{r}_{P/C}) \times \underline{f}_{P\to Q} =$$

$$= \underline{r}_{Q/P} \times \underline{f}_{P\to Q} = 0$$

Then each couple of internal moments is balanced then $\underline{M}^{INT} = 0$

NOTE $\frac{d_o}{dt} I_c(\underline{\omega}_{b/o})$ can be computed in frame <b> _ This is convenient since $\frac{d_b}{dt} \underline{r}_{P/c} \equiv 0$

Then

$$\frac{d_o}{dt} I_c(\underline{\omega}_{b/o}) \overset{(*)}{=} \frac{d_b}{dt} I_c(\underline{\omega}_{b/o}) + \underline{\omega}_{b/o} \times I_c(\underline{\omega}_{b/o})$$

$$= \frac{d_b}{dt} \int_V \underline{r}_{P/c} \times [\underline{\omega}_{b/o} \times \underline{r}_{P/c}] \rho \, dV + \underline{\omega}_{b/o} \times I_c(\underline{\omega}_{b/o})$$

$$= \int_V \underline{r}_{P/c} \times \Big[\underbrace{\frac{d_b}{dt} \underline{\omega}_{b/o}}_{\frac{d_b}{dt}\underline{\omega}_{b/o} = \frac{d_o}{dt}\underline{\omega}_{b/o} \overset{(**)}{=} \dot{\underline{\omega}}_{b/o}} \times \underline{r}_{P/c}\Big] \rho \, dV + \underline{\omega}_{b/o} \times I_c(\underline{\omega}_{b/o})$$

Then

$$\frac{d_o}{dt} I_c(\underline{\omega}_{b/o}) = I_c(\dot{\underline{\omega}}_{b/o}) + \underline{\omega}_{b/o} \times I_c(\underline{\omega}_{b/o})$$

(**) <o> is inertial

(*) Recall $\frac{d_o}{dt} \underline{u} = \frac{d_b}{dt} \underline{u} + \underline{\omega}_{b/o} \times \underline{u}$

On the base of the previous notes eq. (✱) on page 16/ can be re-written as

$$\underline{\underline{I}}_c(\underline{\dot{\omega}}_{b/o}) + \underline{\omega}_{b/o} \times \underline{\underline{I}}_c(\underline{\omega}_{b/o}) = \underline{M}^{EXT} \longleftarrow \text{II dynamic cardinal eq.}$$

$$\underline{0} = \underline{M}^{EXT} \longleftarrow \text{II static cardinal eq.}$$

NOTE $\underline{M}^{EXT}$ is the total contribution of external moments acting on the body $\mathcal{B}$

NOTE The static II cardinal eq. refers to the $\mathcal{B}$ equilibrium state corresponding to $\underline{\omega}_{b/o} \equiv \underline{\dot{\omega}}_{b/o} \equiv \underline{0}$

For the dynamic case the II cardinal eq can be projected onto frame <o>, however recall that the matrix of inertia is usually known in a body fixed ref. frame. The eq. has the form

$${}^{o}\underline{\underline{I}}_c \, {}^{o}\underline{\dot{\omega}}_{b/o} + [{}^{o}\underline{\omega}_{b/o}\times]\, {}^{o}\underline{\underline{I}}_c \, {}^{o}\underline{\omega}_{b/o} = {}^{o}\underline{M}^{EXT}$$

$$({}^{o}_{b}R \, {}^{b}\underline{\underline{I}}_c \, {}^{b}_{o}R) \frac{d}{dt} {}^{o}\underline{\omega}_{b/o} + [{}^{o}\underline{\omega}_{b/o}\times]\, {}^{o}_{b}R \, {}^{b}\underline{\underline{I}}_c \, {}^{b}_{o}R \, {}^{o}\underline{\omega}_{b/o} = {}^{o}\underline{M}^{EXT}$$

to be coupled with

$${}^{o}_{b}\dot{R} = [{}^{o}\underline{\omega}_{b/o}\times]\, {}^{o}_{b}R$$

By projecting the equation of frame $<b>$ we have

$$ {}^bI_c\, {}^b(\dot{\underline{\omega}}_{b/0}) + [{}^b\underline{\omega}_{b/0}\times]\, {}^bI_c\, {}^b\underline{\omega}_{b/0} = {}^b\underline{M}^{EXT} $$

$$ {}^b\left(\frac{d_0}{dt}\underline{\omega}_{b/0}\right) = {}^b\left(\frac{d_b}{dt}\underline{\omega}_{b/0}\right) = \frac{d}{dt}\,{}^b\underline{\omega}_{b/0} $$

Then we have

$$ {}^bI_c \frac{d}{dt}\,{}^b\underline{\omega}_{b/0} + [{}^b\underline{\omega}_{b/0}\times]\, {}^bI_c\, {}^b\underline{\omega}_{b/0} = {}^b\underline{M}^{EXT} $$

to be coupled with

$$ {}^a_b\dot{R} = {}^a_bR\,[{}^b\underline{\omega}_{b/0}] $$

NOTE The II cardinal eq. projected is a set of first order diff. eq.s coupled with a set of first order diff. eq.s (projected Poisson formulas). In both cases since the inertia matrix is positive definite it is invertible then the eq.s have the form

$$ \begin{cases} \dot{\underline{x}} = g_0(\underline{x}, {}^a_bR, {}^0\underline{M}^{EXT}) \\ {}^a_b\dot{R} = [\underline{x}\times]\,{}^a_bR \end{cases} \quad \text{or} \quad \begin{cases} \dot{\underline{x}} = g_b(\underline{x}, {}^a_bR, {}^b\underline{M}^{EXT}) \\ {}^a_b\dot{R} = {}^a_bR\,[\underline{x}\times] \end{cases} $$